

# Muhammad Umar Yaksambi

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## EDUCATION

### RV College of Engineering

BE, Computer Science and Engineering (Data Sciences) – CGPA: 9.12/10

Bengaluru, KA

Expected May 2027

## EXPERIENCE

### HPE CPP3 Intern

Hewlett Packard Enterprise

March 2026 – Present

Remote

- Built a hybrid defect-mapping engine combining a 50/50 lexical and semantic scoring model with BAAI/bge-small-en-v1.5 embeddings to link CI failures to Jira issues, extending it with a 0.90 cosine-similarity duplicate-detection layer within 7-day proximity windows
- Optimized concurrent read/write database performance by implementing SQLite Write-Ahead Logging (WAL) and a 32MB cache extension, eliminating dashboard lockups during active pipeline ingestion
- Implemented a fault-tolerant REST API integration layer with idempotent write-backs and exponential backoff retry logic to safely interact with Atlassian's systems under rate limits

### Product Developer Intern

SpikedAI

June 2025 – March 2026

Remote

- Architected event-driven backend services using FastAPI and Node.js, achieving a sustained 500 RPS throughput to support 500+ concurrent users for a sales meeting intelligence platform
- Optimized RAG and GraphQL pipelines to deliver a 750ms end-to-end response time against an indexed knowledge base of 100M+ tokens, scaling system throughput to 1 million tokens per minute
- Engineered multi-cloud infrastructure across AWS and GCP, orchestrating CI/CD pipelines via GitHub Actions and Nginx load balancing to ensure 99.9% service uptime in production
- Built responsive user interfaces leveraging React, TypeScript, and Tailwind CSS, integrating Redis caching to accelerate the delivery of actionable sales insights generated from Google Meet transcripts

### PRISM Research Intern

Samsung R&D Institute

May 2025 – January 2026

Bengaluru, KA

- Built a real-time volumetric rendering pipeline for a 3D Gaussian Splatting PLY viewer using the Filament engine, achieving 30 FPS at 1080p on Dimensity 9000 mobile hardware with dynamic lighting support, replacing computationally heavy traditional ray-marching
- Optimized spherical harmonics processing and matc material compilation to render large-scale scenes comprising over 3 million Gaussians on mobile hardware

## PROJECTS

### Thrive Learning Platform | Next.js, TypeScript, Supabase, PostgreSQL, OpenAI, Gemini API, PWA

- Engineered an offline-first educational PWA with Next.js and Supabase, integrating OpenAI and Gemini to deliver dynamic curricula, 3D visuals, and a context-aware tutor with local caching and RBAC

### Adaptive-ECC Watermarking for AI Images | Python, OpenCV, Cryptography

- Developed an invisible, adaptive watermarking algorithm for AI images using Error Correction Codes (ECC) that scales intensity by local complexity to protect intellectual property while preserving visual fidelity
- Currently Under Review: *Multimedia Tools and Applications (MTAP)*, Springer

### MedQuery | React, TypeScript, Python, FastAPI, LLMs, Tailwind CSS

- Architected a full-stack clinical platform using React and FastAPI, integrating LLMs to enable healthcare professionals to query the MIMIC-IV dataset via natural language with secure RBAC and interactive analytics

## LEADERSHIP AND ACTIVITIES

### Vice President

The Coding Club of RVCE

February 2025 – Present

Bengaluru, KA

- Led a community of **300+ students** as Vice President of the Coding Club at RVCE, organizing **15+ technical events**, including hackathons, workshops, and coding competitions that engaged **1,000+ participants**.
- Mentored students in software development, AI, and competitive programming while collaborating with faculty, student leaders, and industry partners to strengthen the campus developer community.

## SKILLS

**Languages:** Python, C++, TypeScript, SQL, Embedded C

**Frameworks & Libraries:** PyTorch, TensorFlow, FastAPI, Node.js, React, Next.js, LangChain, GraphQL

**Cloud & DevOps:** AWS, GCP, Kubernetes, Docker, GitHub Actions, Nginx

**Databases:** PostgreSQL, MongoDB, Redis, Qdrant, ChromaDB